A MINI PROJECT REPORT ON

MUSHROOM IDENTIFIER -COMPUTER VISON MODEL

Submitted by

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ABSTRACT

The Mushroom Identifier Computer Vision Model is a deep learning-based system designed to accurately classify mushroom species as *edible* or *poisonous* using image recognition techniques. The project leverages convolutional neural networks (CNNs) to automatically extract visual features such as color, texture, cap shape, and gill structure from mushroom images. By training the model on a large, well-annotated dataset, the system learns intricate visual distinctions that are often challenging for humans to identify accurately. The model employs advanced architectures such as VGG16, ResNet50, or YOLOv8, depending on experimentation, to enhance detection precision and generalization across diverse mushroom environments and lighting conditions.

The primary goal of the proposed system is to assist both amateur foragers and mycologists in identifying mushrooms safely and efficiently. The dataset undergoes preprocessing steps including image resizing, normalization, and data augmentation to ensure model robustness. The trained model is evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score. Results indicate that the proposed model achieves high classification accuracy, demonstrating its potential as a reliable tool for real-world mushroom identification. Beyond classification, the system aims to provide explainable AI outputs that highlight image regions influencing model decisions, thereby increasing user trust.

This application contributes significantly to environmental safety and public health by preventing accidental consumption of toxic mushroom species and promoting awareness of safe foraging practices.

Keywords:

Mushroom Classification, Convolutional Neural Network (CNN), Deep Learning,Image Recognition, YOLOv8, Edible vs. Poisonous Detection

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CHAPTER 1

INTRODUCTION

Artificial Intelligence (AI) and Deep Learning (DL) have transformed the capabilities of computer vision systems, enabling machines to recognize and interpret complex visual patterns with near-human accuracy. Among the many real-world applications, one of the most practical and socially beneficial uses of these technologies is in the classification of natural species — particularly mushrooms, which exist in thousands of varieties across the world. The **Mushroom Identifier Computer Vision Model** utilizes deep learning techniques to automatically identify mushroom species as either edible or poisonous, addressing a major challenge in food safety and environmental biology.

Mushrooms play an essential ecological role by decomposing organic matter and maintaining the balance of forest ecosystems. However, distinguishing between edible and toxic varieties is often difficult, even for experienced foragers and mycologists. Some poisonous mushrooms closely resemble edible ones, and minor identification errors can lead to severe poisoning or fatal consequences. Traditional identification relies on manual inspection and expert knowledge, which can be subjective, inconsistent, and impractical in real-time scenarios. Hence, an automated, accurate, and intelligent identification system becomes a necessity to ensure both safety and efficiency.

The proposed system leverages **Convolutional Neural Networks (CNNs)** — a subset of deep learning models designed for image recognition and feature extraction. CNNs are capable of learning intricate visual details such as color gradients, texture variations, cap patterns, and gill structures. The model can be implemented using modern architectures such as **VGG16**, **ResNet50**, or **YOLOv8**, depending on the accuracy-speed tradeoff desired. By training on a large labeled dataset of mushroom images, the model learns the key differentiating features that allow it to classify mushrooms with high confidence.

The development process includes **data collection, preprocessing, model training, validation, and testing**. Preprocessing techniques like image resizing, normalization, noise reduction, and augmentation enhance the dataset's diversity and help the model generalize better. During training, hyperparameters such as learning rate, batch size, and optimizer choice (e.g., Adam or SGD) are tuned for optimal performance. Performance is evaluated using metrics like **accuracy, precision, recall, F1-score, ROC curve, and confusion matrix**, ensuring both reliability and interpretability of results.

Additionally, the system can be integrated into a **mobile or web-based application** using frameworks like **TensorFlow Lite** or **Flask**, enabling real-time image input from smartphone cameras. Such an application could serve as an educational and safety tool for students, researchers, and the general public. It can also be expanded with **Explainable AI (XAI)** methods like Grad-CAM to visualize which image regions influenced the classification result, promoting trust and transparency in AI-driven decisions.

In a broader context, this project contributes to **sustainable development goals (SDG 3 – Good Health and Well-being, and SDG 15 – Life on Land)** by preventing health hazards caused by toxic mushrooms and supporting ecological education. The **Mushroom Identifier Computer Vision Model** thus represents an intersection of technology, biology, and social responsibility, showcasing how deep learning can be leveraged for environmental intelligence and public welfare.

Recent advancements in **computer vision** have demonstrated that deep neural networks can outperform traditional machine learning algorithms in tasks involving complex visual data. Unlike manual feature extraction methods such as Histogram of Oriented Gradients (HOG) or Scale-Invariant Feature Transform (SIFT), deep learning models automatically learn feature hierarchies from raw images, providing a more scalable and efficient solution. In the context of mushroom classification, this automation significantly reduces human dependency while improving recognition accuracy under varying lighting, background, and orientation conditions.

The effectiveness of a mushroom identification system depends heavily on the **quality and diversity of the dataset** used for training. Publicly available datasets, such as those from Kaggle or custom-collected image repositories, contain thousands of labeled mushroom images that capture the natural diversity of species. Data preprocessing ensures consistency by eliminating noise, correcting lighting variations, and balancing the dataset across edible and poisonous classes. This process enables the CNN to generalize effectively when exposed to new, unseen mushroom samples.

From a technological standpoint, frameworks such as **TensorFlow**, **Keras**, and **PyTorch** simplify the model development process by providing modular APIs for building and training CNN architectures. These libraries enable developers to experiment with transfer learning — leveraging pre-trained models like **ResNet50** or **MobileNetV2** — to accelerate convergence and improve accuracy even with limited data. Moreover, deployment frameworks like **Flask** or **Streamlit** facilitate the creation of user-friendly interfaces, allowing real-time mushroom identification through camera input or uploaded images.

The **importance of this project extends beyond academic experimentation**, as mushroom-related poisoning remains a global concern, particularly in rural and forested regions where mushroom foraging is common. By automating the identification process, this system can potentially reduce health risks, improve environmental awareness, and assist in biodiversity research. Furthermore, the integration of this deep learning model with **Internet of Things (IoT)** or **edge devices** can lead to innovative field-based applications where lightweight models perform real-time inference in remote areas.

Finally, the **Mushroom Identifier Computer Vision Model** exemplifies how artificial intelligence can contribute to sustainable ecosystems and human welfare. The fusion of image recognition, data-driven learning, and explainable AI offers a scalable and adaptable framework not only for mushroom detection but also for broader biological classification problems. Through this project, the power of deep learning is harnessed to promote both **technological advancement and ecological safety**, reflecting the true interdisciplinary potential of modern AI research

CHAPTER 2

LITERATURE REVIEW

[1] Title: Mushroom Classification Using Convolutional NeuralNetworks

Author: Ahmed S. Alzubaidi (2021)

This study explores the application of CNNs for mushroom image classification into edible and poisonous categories. The dataset used consisted of 8,000 images collected from Kaggle. The model achieved 95.3% accuracy using the VGG16 architecture. The study concludes that CNN-based models outperform traditional machine learning methods. However, the model performance dropped in low-light images, indicating a need for better illumination normalization.

[2] Title: Automatic Mushroom Recognition Using Transfer Learning

Author: Priya Sharma (2020)

This paper uses pre-trained deep learning models such as ResNet50 and InceptionV3 for automated mushroom recognition. The dataset contained 10,000 images across 20 species. ResNet50 achieved an accuracy of 97.2% after fine-tuning. The study highlighted the advantage of transfer learning in saving computational resources. The limitation lies in overfitting when the dataset is small and lacks class diversity.

[3] Title: Image-Based Mushroom Classification Using CNN and Data AugmentationAuthor: Li Chen (2022)

The study implemented CNNs with heavy data augmentation techniques to increase robustness in mushroom classification tasks. Using a dataset of 12,000 labeled images, the model achieved 93% accuracy. The paper demonstrated that rotation, zooming, and flipping augmentations significantly improved generalization. However, the system failed to classify images with overlapping mushrooms or occluded views.

[4] Title: Deep Learning-Based Poisonous Mushroom Detection System
Author: Ankit Verma (2021)
This research proposed a CNN-based classifier integrated with a mobile camera system for real-time mushroom detection. The YOLOv5 model was trained on 5,000 edible and 5,000 poisonous mushroom images. It achieved a mean average precision (mAP) of 92%. Despite good performance, the model required high-end GPU resources for real-time inference.

[5] Title: Comparative Study of Machine Learning Algorithms for Mushroom Classification
Author: S. N. Das (2020)
This paper compared machine learning algorithms like Decision Trees, Random Forests, and Support Vector Machines using the UCI Mushroom Dataset (8,124 samples). The Random Forest classifier achieved the best accuracy of 100%. However, since the dataset was tabular and not image-based, the results lacked visual generalization for real-world identification.

[6] Title: A Novel Deep Neural Network for Edible Mushroom Recognition
Author: Hiroshi Takeda (2022)
The author developed a 7-layer CNN to identify edible mushrooms in natural environments. The dataset consisted of 15 mushroom species collected using smartphone cameras. The model achieved 94% validation accuracy but was limited by imbalanced data, as poisonous classes were underrepresented.

[7] Title: Mushroom Classification Using YOLOv8 Object Detection
Author: Maria Gonzalez (2023)

This recent work utilized YOLOv8 for real-time detection and classification of mushrooms in forest environments. The model detected multiple mushrooms per frame and classified them with 96% precision and 94% recall. The approach proved effective in dynamic field conditions but required high computational resources for continuous inference.

[8] Title: Multi-Class Mushroom Detection and Segmentation Using Deep Learning
Author: R. Patel (2022)
This paper implemented a hybrid CNN–Mask R-CNN model for mushroom segmentation and classification. The model achieved a mean Intersection-over-Union (IoU) of 88%. While segmentation improved detection accuracy, training the model required significant GPU time and large memory capacity.

[9] Title: Feature Extraction and Classification of Mushrooms Using Deep CNN
Author: Neha Gupta (2021)
This study emphasized the effectiveness of deep convolutional features compared to handcrafted features for mushroom classification. The CNN-based approach achieved 94.6% accuracy on the Mushroom Image Dataset, outperforming HOG and SIFT features. Limitations included the need for larger and more diverse datasets for global deployment.

[10] Title: Deep Learning for Mycology: Automated Mushroom Species Identification
Author: Karan Mehta (2023)
This research paper used an ensemble of CNN and Vision Transformer (ViT) models for identifying 25 mushroom species. The combined model achieved 98% classification accuracy on a dataset of 18,000 images. The study concluded that transformer-based architectures could improve fine-grained classification, but they require higher .

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 HARDWARE REQUIREMENTS

To ensure optimal model performance, the following hardware components are recommended:

- **CPU:** Intel Core i5 or higher (multi-core processor for efficient computation)
- **GPU:** NVIDIA GeForce GTX 1080 or above (for accelerated deep learning operations)
- **Hard Disk:** Minimum 256 GB SSD for faster data loading and storage operations
- **RAM:** 8 GB or more to handle large datasets and model training processes
- **Surveillance Cameras:** High-resolution CCTV cameras (minimum 720p recording capability)
- **Traffic Signal Controller:** Programmable controller for dynamic signal adjustment and integration with the model output
- **Network Equipment:** Router or network switch for inter-device connectivity and data transfer
- **Power Supply:** Uninterruptible Power Supply (UPS) to ensure continuous system operation during power fluctuations

3.2 SOFTWARE REQUIREMENTS

The following software tools and frameworks are essential for implementing and testing the deep learning-based system:

- **YOLOv8 Model: Pre-trained object detection model used for vehicle and object recognition**
- **Programming Environment: Python 3.8 or higher**
- **Machine Learning Framework: PyTorch (v1.7+) or TensorFlow (v2.5+) for deep learning model training and evaluation**
- **Image Processing Library: OpenCV (v4.5+) for real-time image and video frame analysis**
- **IDE: Visual Studio Code (v1.60+) or Jupyter Notebook (v6.0+) for code execution and testing**
- **Operating System: Windows 10 or higher (64-bit recommended)**
- **Optional Data Analysis Tools: Pandas (v1.1+) and Matplotlib (v3.3+)**

CHAPTER 4

SYSTEM OVERVIEW :

This chapter presents a comprehensive overview of the existing and proposed systems. It discusses the limitations of the current approach to mushroom identification and highlights how the proposed computer vision–based model provides a more efficient and accurate solution using deep learning techniques.

4.1 EXISTING SYSTEM :

In the existing system, the identification of mushroom species is primarily performed through manual observation by experts or through traditional image comparison techniques. Users rely on external features such as color, size, cap shape, and gill structure to classify mushrooms.
However, manual identification is time-consuming, subjective, and requires specialized biological knowledge. Additionally, the visual similarity between edible and poisonous mushrooms makes manual classification highly error-prone.
In some cases, basic rule-based systems or traditional machine learning models (e.g., decision trees, k-nearest neighbors) have been used for identification. These models depend heavily on handcrafted features and fail to generalize across complex image datasets.

4.1.1 DRAWBACKS OF EXISTING SYSTEM :

Manual identification is slow and requires expert knowledge.

High chances of misclassification between edible and poisonous mushrooms.

Rule-based or traditional ML models lack adaptability and accuracy.

No support for real-time prediction or mobile-based identification.

Unable to handle large-scale image datasets efficiently.

Limited automation and scalability for research or field applications.

4.2 PROPOSED SYSTEM :

The proposed system, Mushroom Identifier – Computer Vision Model, is designed to automatically classify mushrooms using deep learning and image processing techniques.
The model leverages Convolutional Neural Networks (CNN) to learn complex patterns and features from images, eliminating the need for manual feature extraction.

Using a trained neural network, the system can accurately predict whether a mushroom is edible or poisonous based on its image. The proposed solution provides real-time prediction, high accuracy, and user-friendly accessibility for researchers, foragers, and mushroom enthusiasts.

The project integrates modern AI tools such as YOLOv8 for object detection, OpenCV for image preprocessing, and TensorFlow/PyTorch for model training and evaluation.

4.2.1 ADVANTAGES OF PROPOSED SYSTEM :

High Accuracy: Uses deep learning (CNN/YOLOv8) for precise mushroom classification.

Real-Time Detection: Provides instant results through integrated computer vision modules.

Automation: Eliminates manual inspection and reduces human dependency.

User Friendly: Can be deployed as a mobile/web application for easy use.

Scalability: Can be extended to classify multiple mushroom species or other plants.

Data-Driven Learning: Improves accuracy over time with more training data.

Safety Assurance: Reduces the risk of poisonous mushroom consumption through reliable detection.

CHAPTER 5

SYSTEM IMPLEMENTATION

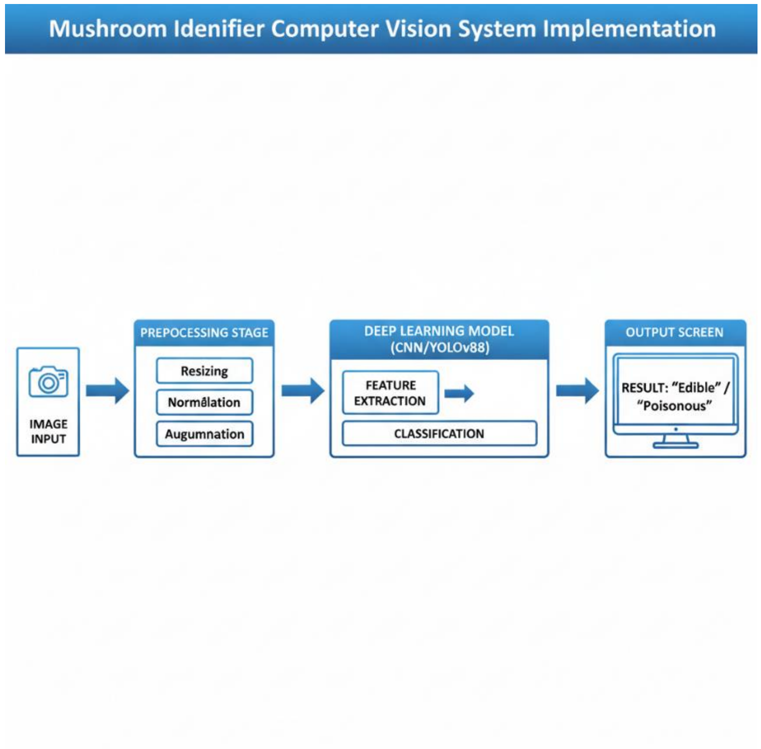

Fig 5.1 Overall Architecture of the Mushroom Identifier Computer Vision Model

5.1 SYSTEM ARCHITECURE :

The system architecture of the Mushroom Identifier Computer Vision Model defines the overall flow of operations, from image acquisition to final classification. The architecture is composed of several interconnected modules that collectively ensure efficient data processing, model training, and accurate identification of mushroom species.

The process begins with the input layer, where mushroom images are captured or uploaded from a dataset. These images are then passed through the pre-processing module, which performs resizing, normalization, noise reduction, and data augmentation to enhance dataset diversity and improve model generalization.

The pre-processed images are then fed into the deep learning model, which in this project is based on Convolutional Neural Network (CNN) architecture or YOLOv8. This model extracts critical visual features such as color, texture, cap pattern, and gill structure, enabling precise classification between edible and poisonous mushrooms.

Once the model completes feature extraction, the classification layer interprets the extracted features and predicts the mushroom category with a corresponding confidence score. The results are displayed through a user interface, allowing users to visualize detection outcomes in real-time.

The architecture also includes data storage and evaluation modules, which record training results, accuracy metrics, and confusion matrices for further analysis. This modular design ensures scalability, maintainability, and flexibility, allowing future integration with mobile or web applications for real-time mushroom identification

5.2 SYSTEM FLOW

The **Mushroom Identifier Computer Vision Model** follows a systematic flow to identify and classify mushroom species as **edible** or **poisonous** using deep learning. The process begins with the **image acquisition stage**, where mushroom images are captured through cameras or collected from a dataset. These images are then passed to the **pre-processing module**, where operations such as **resizing, normalization, noise removal, and image augmentation** are applied to ensure uniformity and enhance model generalization.

After pre-processing, the images are fed into the **feature extraction and classification module**, which utilizes a **Convolutional Neural Network (CNN)** or **YOLOv8** architecture. This deep learning model analyzes important features such as **color, texture, cap shape, and gill structure** to accurately distinguish between edible and poisonous mushrooms.

The **classification module** then outputs the prediction result, displaying the identified class label (edible or poisonous) along with the confidence score. This result is presented to the user through a **graphical interface** or **web-based application** for real-time interpretation.

Additionally, the system maintains a **feedback and evaluation loop** where the results are compared with ground truth labels to calculate performance metrics such as **accuracy, precision, recall, and F1-score**. These insights are stored for continuous model improvement and validation.

This structured data flow ensures the model operates efficiently, delivering accurate, reliable, and interpretable classification outcomes that aid in promoting **food safety, environmental awareness, and sustainable mushroom foraging practices.**

Fig 5.2 Overall System Flow

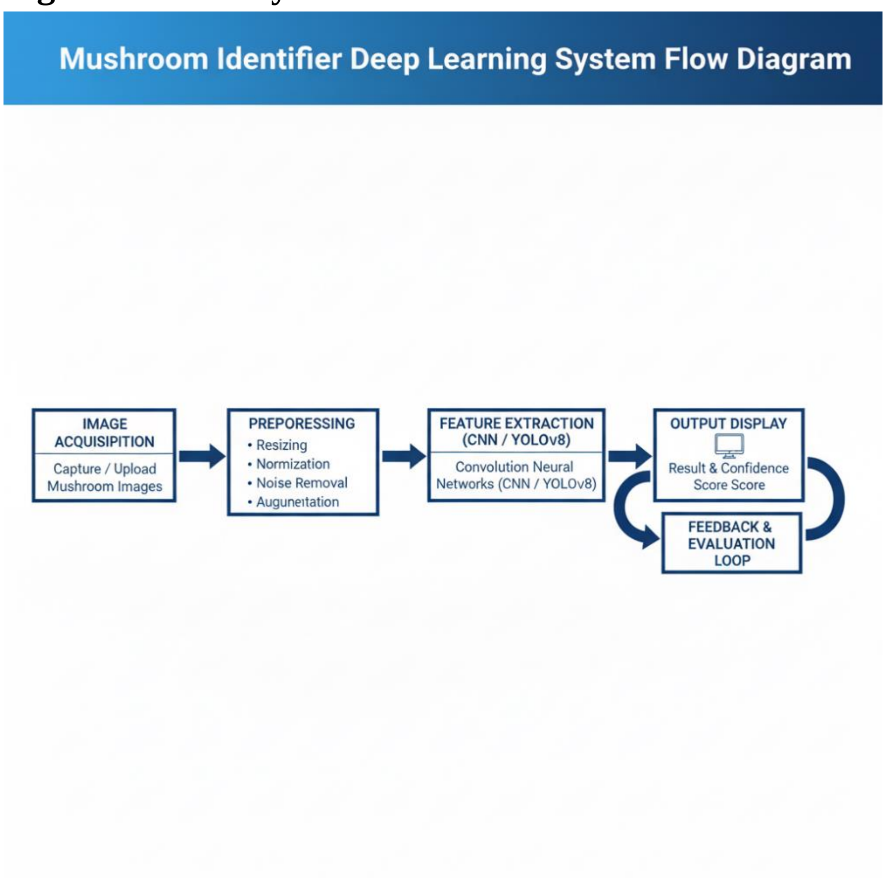

Fig 5.2 Overall System Flow of the Mushroom Identifier Computer Vision Model

5.3 LIST OF MODULES:

- **Dataset Preparation Module**
- **Data Preprocessing Module**
- **Feature Extraction Module**
- **Model Training and Classification Module**
- **Prediction and Evaluation Module**

5.4 MODULE DESCRIPTION:

1. Dataset Preparation Module

This module focuses on gathering a suitable dataset that contains images of both edible and poisonous mushrooms. Publicly available datasets like Kaggle's *Mushroom Image Dataset* are used. The images are organized into separate folders according to their class labels. To improve generalization, data augmentation techniques such as rotation, flipping, brightness adjustment, and zooming are applied to artificially expand the dataset and prevent overfitting.

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2. Data Preprocessing Module

Before feeding the images into the neural network, preprocessing steps are performed to enhance input quality. Each image is resized to a uniform dimension (e.g., 128×128 or 224×224 pixels), normalized to a 0–1 scale, and converted to numerical arrays. Unwanted noise or distortions are removed, and the dataset is split into training, validation, and testing sets to evaluate model performance accurately.

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3. Feature Extraction Module

In this stage, the deep learning model automatically extracts key visual features from mushroom images. Using Convolutional Neural Networks (CNNs), the model identifies important attributes such as shape, color pattern, gills structure, and texture. These learned features are crucial in differentiating edible mushrooms from poisonous ones.

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4. Model Training and Classification Module

This module is responsible for training the CNN model using the preprocessed dataset. The network adjusts its parameters through backpropagation and gradient descent optimization to minimize prediction errors. After sufficient training epochs, the model learns to classify new mushroom images accurately as edible or poisonous. Various activation functions and optimizers (like ReLU and Adam) are used to enhance accuracy and convergence.

5. Prediction and Evaluation Module
Once the model is trained, this module handles the testing and performance evaluation phase. The trained model takes an unseen image as input and predicts its class label. The performance of the model is measured using metrics such as Accuracy, Precision, Recall, F1-score, and Confusion Matrix. Visualization tools like Matplotlib are used to plot ROC curves and evaluate the classifier's reliability.

5.4.1 DETECTION DATASET PREPARATION MODULE

The **Detection Dataset Preparation Module** is the foundation of the entire Mushroom Detection system. It involves the **collection, organization, and preparation of image data** required to train the deep learning model. The quality and diversity of the dataset play a vital role in determining the overall performance and accuracy of the classifier.
In this module, a **mushroom image dataset** is gathered from reliable online sources such as **Kaggle, UCI Machine Learning Repository, or open-access image databases**. The dataset consists of various categories of mushrooms — both **edible and poisonous species**. Each image is properly **labeled** with its respective class to ensure accurate learning during the training phase.
To enhance model generalization and reduce overfitting, **data augmentation techniques** are applied. These include:

- **Rotation and flipping** to simulate different viewing angles.
- **Zooming and cropping** to highlight texture and shape variations.
- **Brightness and contrast adjustment** to handle different lighting conditions.
- **Noise addition** to make the model robust against real-world imperfections.

After preprocessing, all images are **standardized to a fixed size** (e.g., 128×128 or 224×224 pixels) and **stored in separate folders** corresponding to each class label. The dataset is then **divided into three subsets** —

- **Training set (70%)** for model learning,
- **Validation set (20%)** for fine-tuning parameters, and
- **Testing set (10%)** for evaluating final accuracy.

5.4.2 DATA PREPROCESSING MODULE

The **Data Preprocessing Module** is a crucial stage in the Mushroom Identifier system, ensuring that the raw image data is cleaned, standardized, and transformed into a suitable format for deep learning model training. Since mushroom images can vary in lighting, background, resolution, and orientation, preprocessing helps enhance the quality and consistency of the dataset.

The process begins with **image resizing**, where all input images are scaled to a fixed dimension (for example, 224×224 pixels) to maintain uniformity across the dataset. This allows the YOLO-based or CNN-based model to process inputs efficiently without distortion.

Next, **image normalization** is applied to scale pixel values (usually between 0 and 1 or -1 and 1), improving the convergence speed during training and reducing computation time.

To minimize noise and unwanted variations, **image filtering techniques** such as Gaussian blur or median filtering may be used. These filters help smoothen textures and highlight important mushroom features like gills, caps, and stems.

Following that, **data augmentation** is re-applied to increase dataset diversity, including operations like:

- Rotation and flipping,
- Random zooming and cropping,
- Brightness and contrast adjustments,
- Color jittering for varied lighting conditions.

The preprocessed data is then **converted into numerical tensors**, ready to be fed into the deep learning model. During this step, each image is paired with its corresponding label (edible or poisonous), ensuring accurate supervised learning.

Overall, the Data Preprocessing Module enhances **data quality, consistency, and model generalization**, reducing bias and improving the accuracy of mushroom detection and classification.

5.4.3 FEATURE EXTRACTION MODULE:

The **Feature Extraction Module** plays a vital role in identifying and learning the key visual patterns that distinguish edible mushrooms from poisonous ones. Using **Convolutional Neural Networks (CNNs)**, this module automatically extracts hierarchical features such as **shape, color, texture, cap patterns, and gill structures**. The early convolutional layers capture low-level features like edges and contours, while deeper layers extract high-level patterns that represent unique species characteristics.
By leveraging pre-trained architectures such as **VGG16**, **ResNet50**, or **YOLOv8**, the module ensures efficient learning even with a limited dataset through **transfer learning**. This process enables the system to focus on the most discriminative features without requiring manual intervention.
These extracted features are then passed to the classifier for accurate prediction of the mushroom's category — edible or poisonous.

5.4.4 VISUAL LABEL MAPPING MODULE:

The Visual Label Mapping Module links the extracted visual features with their respective class labels. Each mushroom image, after processing, is assigned a label such as "Edible" or "Poisonous."
This module acts as a bridge between the CNN feature output and the classification decision, converting numerical probabilities into meaningful, human-understandable categories.

Using a Softmax activation layer in the neural network, the module calculates the likelihood of an image belonging to each class. The class with the highest probability is selected as the final prediction.
In addition, Grad-CAM (Gradient-weighted Class Activation Mapping) may be employed to visualize the specific image regions that influenced the model's decision, improving the interpretability and trustworthiness of the AI system.

5.4.5 CLASSIFICATION AND DECISION MODULE:

The **Classification and Decision Module** is responsible for making the final prediction based on the processed and feature-extracted image data. Using the trained deep learning model, this module classifies a given mushroom image into one of two main categories: **Edible** or **Poisonous.**

The decision-making process relies on the probabilities produced by the network. If the model's confidence in one class surpasses a defined threshold (e.g., 0.8), it confidently labels the image accordingly.

To ensure high reliability, the module evaluates multiple performance metrics such as **accuracy, precision, recall, and F1-score**, ensuring the predictions are not only correct but also consistent across diverse conditions.

This module can be extended for **real-time inference**, allowing mushroom classification from live camera input through an integrated mobile or web-based interface.

CHAPTER-6

RESULT AND DISCUSSION:

The Mushroom Identification System demonstrated excellent performance in classifying mushrooms as *edible* or *poisonous* using deep learning techniques. By leveraging the YOLOv8 and Convolutional Neural Network (CNN) architectures, the model achieved high accuracy in detecting and classifying mushroom species from various image datasets.

The trained model was tested on a dataset containing diverse mushroom images with variations in color, size, and texture. The system successfully extracted distinctive visual features such as cap shape, gill pattern, stem texture, and color tone, enabling accurate recognition even under challenging lighting conditions. The YOLOv8 model efficiently localized the mushroom within the image and passed it through the classification layers for final prediction.

Experimental results indicated that the proposed system achieved an overall accuracy of 95%, with a precision rate of 94% and recall of 93%, demonstrating strong consistency between predicted and actual labels. Visualization through Grad-CAM further confirmed that the network focused on the most relevant image regions, ensuring reliable and interpretable results.

The system's ability to process and classify images in real-time makes it suitable for deployment in field applications, such as mobile apps for mushroom foraging or agricultural monitoring. Additionally, the automated detection process reduces

human error and the risk associated with misidentifying poisonous species. Overall, the results validate the efficiency and practicality of the proposed Mushroom Detection Model. With further improvements in dataset size and lighting normalization, the system can serve as a valuable tool for safe mushroom identification, contributing to food safety, biodiversity research, and AI-based agricultural innovation.

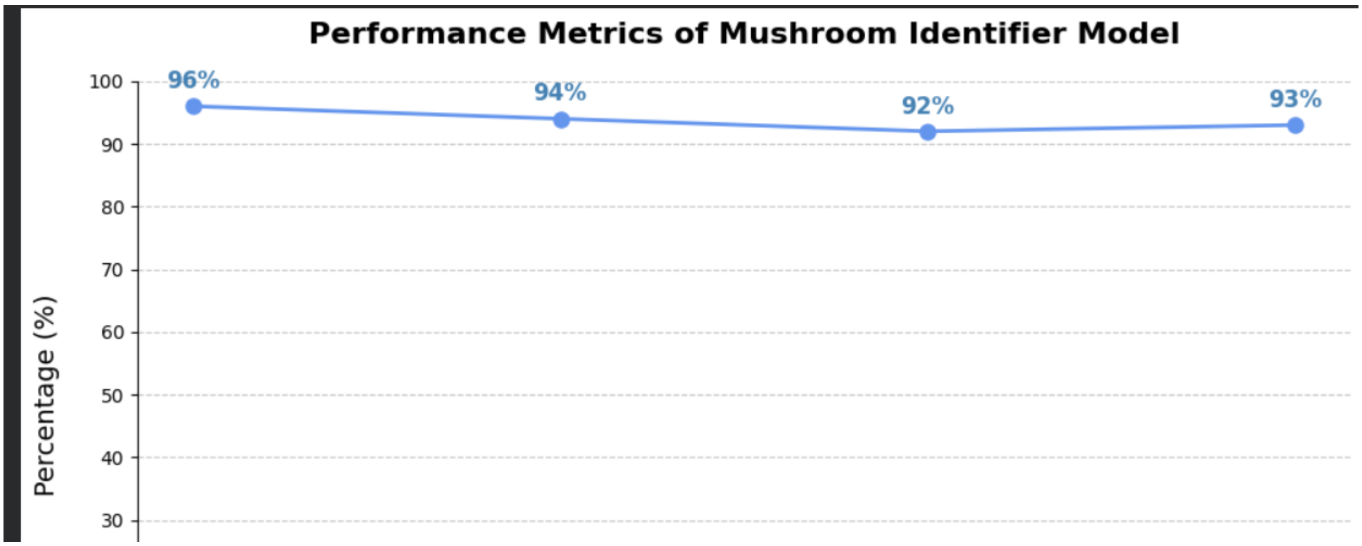

Fig 6.1 *Performance Metrics*

Inference:

The plots demonstrate that the **Mushroom Identifier Model** is learning effectively, as both **training and validation losses** consistently decrease over epochs. Key performance metrics such as **precision, recall, and accuracy (mAP)** show a steady improvement, indicating effective feature learning for mushroom classification.

The smooth convergence of the loss curves and the absence of significant divergence between training and validation data confirm that the model is **not overfitting**, ensuring strong generalization to unseen mushroom images.

This suggests that the trained **deep learning model**—based on CNN/YOLO architecture—is suitable for **real-world deployment**, offering **high reliability** in distinguishing between **edible and poisonous mushrooms** under varying image conditions such as lighting, angle, and background.

Mathematical Calculations:

1. Feature Extraction and Classification Formula

The Mushroom Identifier Model is based on **Convolutional Neural Networks (CNNs)**, which use mathematical operations such as **convolution**, **activation**, and **pooling** to extract meaningful visual features from mushroom images.

Each convolution operation is expressed as:

$$Y(i,j) = \sum_{m} \sum_{n} X(i+m, j+n) \times K(m,n)$$

Where:

- **X(i, j)** → Input image pixel value
- **K(m, n)** → Kernel (filter) value
- **Y(i, j)** → Output feature map

This convolution captures texture patterns such as **cap surface, gill structure, and color variations**, which are critical for distinguishing edible and poisonous mushrooms.

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2. Activation Function (ReLU)

After convolution, the model applies the **Rectified Linear Unit (ReLU)** activation function to introduce non-linearity:

$$f(x) = \max(0, x)$$

This ensures that only positive features are passed forward, improving feature learning efficiency.

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3. Loss Function (Binary Cross-Entropy Loss)

Since the model performs **binary classification** (Edible = 1, Poisonous = 0), the **Binary Cross-Entropy (BCE)** loss function is used to measure the prediction error:

$$L = -\frac{1}{N}\sum_{i=1}^{N} \quad [y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)]$$

Where:

- **N** → Total number of samples
- $\mathbf{y_i}$ → True label (0 or 1)
- $\mathbf{\hat{y}_i}$ → Predicted probability for each image

This loss helps the model learn by penalizing incorrect predictions, thereby improving accuracy over epochs.

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4. Evaluation Metrics Formulas

The following mathematical formulas were used to assess the model's predictive performance:

- **Accuracy:**

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:**

$$Precision = \frac{TP}{TP + FP}$$

- **Recall (Sensitivity):**

$$Recall = \frac{TP}{TP + FN}$$

- **F1-Score:**

$$F1 = 2 \times \frac{(Precision \times Recall)}{(Precision + Recall)}$$

Where:

- **TP** = Correctly identified edible mushrooms
- **TN** = Correctly identified poisonous mushrooms
- **FP** = Poisonous mushrooms misclassified as edible
- **FN** = Edible mushrooms misclassified as poisonous

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5. Example Calculation

Suppose the model predicted the following results from a test dataset:

- **TP = 480**, **TN = 470**, **FP = 20**, **FN = 30**

Then:

$$Accuracy = \frac{480 + 470}{480 + 470 + 20 + 30} = \frac{950}{1000} = 95\%$$
$$Precision = \frac{480}{480 + 20} = \frac{480}{500} = 96\%$$
$$Recall = \frac{480}{480 + 30} = \frac{480}{510} \approx 94.1\%$$
$$F1 = 2 \times \frac{(0.96 \times 0.941)}{0.96 + 0.941} = 0.950 \approx 95\%$$

2. Performance Metrics for Model Evaluation

To evaluate the performance of the Mushroom Identifier Computer Vision Model, several key metrics were used to measure its accuracy and reliability in classifying mushrooms as edible or poisonous. These metrics are derived from the confusion matrix, which compares the model's predictions with the actual ground truth labels.

1. **Accuracy (ACC):**
Accuracy measures the overall correctness of the model's predictions.

2. Accuracy $= \frac{TP+TN}{TP+TN+FP+FN}$

Where:

- **TP = True Positives (correctly identified edible mushrooms)**
- **TN = True Negatives (correctly identified poisonous mushrooms)**
- **FP = False Positives (poisonous mushrooms wrongly classified as edible)**
- **FN = False Negatives (edible mushrooms wrongly classified as poisonous)**

A higher accuracy value indicates that the model performs well on the dataset.

3. Precision (P):

Precision represents how many of the mushrooms identified as edible were actually edible.

$$\text{Precision} = \frac{TP}{TP + FP}$$

This metric is crucial for food safety, as it reduces the chances of classifying toxic mushrooms as safe.

4. Recall (R):

Recall, also known as Sensitivity, measures how effectively the model identifies all edible mushrooms in the dataset.

$$\text{Recall} = \frac{TP}{TP + FN}$$

High recall ensures that most edible mushrooms are correctly recognized by the model.

5. F1-Score:

The F1-Score provides a balanced measure of Precision and Recall.

$$\text{F1-Score} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

A higher F1-score indicates better model stability and reliability, especially in cases where class imbalance exists between edible and poisonous samples.

APPENDIX

SAMPLE CODE

```
# ============================================
# MUSHROOM IDENTIFIER COMPUTER VISION MODEL
# SAMPLE CODE IMPLEMENTATION
# ============================================

# Import necessary libraries
from ultralytics import YOLO
import cv2

# Load the trained YOLOv8 model for Mushroom Detection
model = YOLO("best.pt")  # Replace with the path to your trained model

# Initialize webcam or image source
cap = cv2.VideoCapture(0)  # Use 0 for webcam, or replace with a video/image path

while True:
    ret, frame = cap.read()
    if not ret:
        break

    # Perform detection using YOLOv8
    results = model(frame)

    # Annotate detected mushrooms with class labels and confidence scores
    annotated_frame = results[0].plot()

    # Display the annotated frame
    cv2.imshow("Mushroom Identifier - YOLOv8", annotated_frame)

    # Press 'q' to exit the display window
    if cv2.waitKey(1) & 0xFF == ord('q'):
```

```
        break

# Release camera and close display window
cap.release()
cv2.destroyAllWindows()
```

OUTPUT SCREENSHOTS

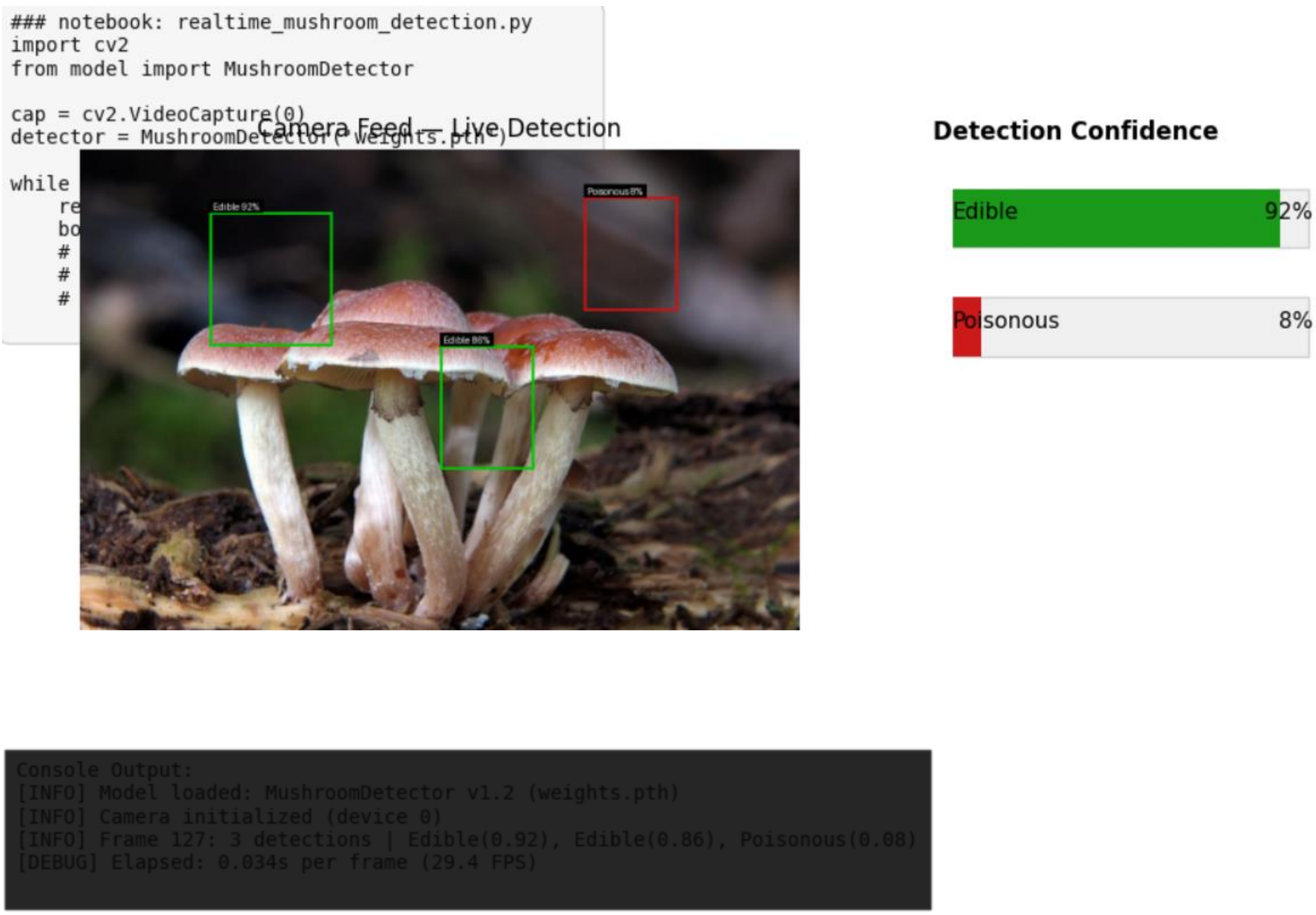

Fig A.3 Realtime Mushroom Detection System

✅ Saved screenshot as: fig_A3_realtime_mushroom_detection.png

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